1. Clinical Question & Core Hypothesis

Clinical Question

Intravesical BCG provokes local immune recruitment but also up-regulates PD-L1; why do some immune-infiltrated (TLS/TIL-rich) urothelial cancers nevertheless fail to respond to BCG? 2) Tier 1 paths used: the chain linking PI3-kinase pathway activation → PD-L1 and the BCG → PD-L1 induction; I represent PI3K activation by PIK3CA (tumor-intrinsic driver) and PD-L1 by CD274 because these directly explain constitutive versus therapy-induced PD-L1. 3) Tier 2/supporting evidence: TMPRSS2/ERG and PSMA (FOLH1) appear in the path as upstream oncogenic/contextual modifiers that could alter immune phenotype or cross-talk between tumor types; they support exploring whether oncogenic programs rewire PD-L1 regulation. 4) Endpoint choice: therapy_response (BCG response) is the most relevant clinical question for this chain, since the path centers on BCG-induced immune modulation and PD-L1-driven immune suppression. 5) Deliberate exclusions: I excluded broader pathway descriptors (e.g., generic 'PI3K pathway' label) in favor of a measurable gene (PIK3CA) and omitted additional exploratory genes (e.g., other immune checkpoints, metabolic regulators) to keep the hypothesis sharply testable and within the 1 imaging + 2 gene constraint?

Working Hypothesis

TLS-rich urothelial carcinomas with tumor-intrinsic PI3K activation driven by PIK3CA alterations sustain constitutive CD274 (PD-L1) expression within the tumor microenvironment, establishing a PD-L1–dominant immunosuppressive niche that dampens BCG-induced immune recruitment and activation. Consequently, TLS-high tumors with PIK3CA-driven PD-L1 upregulation will exhibit poorer clinical response to intravesical BCG compared with TLS-high tumors lacking this PI3K/PD-L1 axis activation. Secondary modifiers—TMPRSS2/ERG fusion status and FOLH1 (PSMA)–related signaling—may further influence PD-L1 levels or TLS maturation and thereby modulate the magnitude of BCG response in TLS-rich tumors. Imaging-feature TLS-like aggregates will serve as a phenotypic readout linking tumor-intrinsic PI3K activity to the immunosuppressive TLS niche and BCG responsiveness.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CD274, PIK3CA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CD274: CD274 because these directly explain constitutive versus therapy-induced PD-L1.

PIK3CA: PIK3CA (tumor-intrinsic driver) and PD-L1 by CD274 because these directly explain constitutive versus therapy-induced PD-L1.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TMPRSS2	Supporting modifier from knowledge graph
Secondary	ERG	Supporting modifier from knowledge graph
Secondary	FOLH1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of BCG-treated cohorts with available sequencing and IHC data to correlate PIK3CA status, PD-L1 expression, TLS density, and response;

Prediction 2: prospective validation with baseline profiling of PIK3CA, CD274, TMPRSS2/ERG, and FOLH1 and TLS imaging;

Prediction 3: mechanistic studies using urothelial models to assess whether PI3K activation drives PD-L1 upregulation and whether PI3K inhibition or PD-L1 blockade rescues BCG-induced immune recruitment;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TMPRSS2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ERG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.